

Kane Pavlovich

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Professional Profile

PhD candidate in computational neuroscience with 4+ years experience building predictive models and extracting insights from complex datasets. Expertise in machine learning (Python), advanced statistics (R), network analysis, and data visualisation with a proven ability to translate technical findings for diverse stakeholders. Passionate about developing data-driven solutions rooted in rigorous computational methods, with experience leading cross-disciplinary projects from conception to implementation.

Work Experience

HEAD TUTOR / GUEST LECTURER | MONASH UNIVERSITY | FEBRUARY 2025 – PRESENT

- Designed and delivered statistics & psychology focused curriculum for 500+ undergraduates, translating complex technical concepts into accessible lectures and tutorials for diverse audiences.
- Led a team of 12 tutors to standardise instruction, optimize grading workflows, and implement feedback systems, demonstrating operational leadership.

TEACHING FELLOW / ASSISTANT LECTURER | MONASH UNIVERSITY | FEBRUARY 2024 – FEBRUARY 2025

- Designed and optimised lecture materials, assessments, and learning tools for psychology courses, ensuring alignment with measurable outcomes (e.g., exam performance, engagement metrics).
- Synthesized feedback from students, faculty, and course directors to identify the biggest gaps in curriculum, leading to the development of new course modules and assignments to maximise students learning.

Education

PHD | JANUARY 2022 - JULY 2025 | MONASH UNIVERSITY

- Developed predictive brain-behavior models using machine learning, network analysis, and statistical modeling, establishing scalable frameworks for high-dimensional neuroscience data.
- Published 3 first author peer-reviewed research in top academic articles, with emphasis on interpretable data visualisation and reproducible methodology to inform future predictive models.
- Presented work at international conferences, communicating data-driven insights to interdisciplinary stakeholders (scientists, engineers, clinicians).
- Co-supervised 2 honours students in statistical methods and engineering, leading them to produce high quality honours theses.

MASTERS OF PSYCHOLOGY | FEBRUARY 2018 - OCTOBER 2019 | UNIVERSITY OF AUCKLAND

- Applied time-series analysis and spectral analysis (e.g., Fourier transforms, wavelet decompositions) to investigate attention-driven oscillatory dynamics in neural data, demonstrating strong signal processing and quantitative research skills.
- Awarded with first-class honours.

Skills

- **Python** for machine learning (Classification & Regression) (Pandas, NumPy, scikit-learn)
- **R** for advanced statistical analyses (Structural Equation modelling) (dplyr, tidyverse, lavaan)
- **MATLAB** for dimensionality reduction, time-series analysis, network/graph analysis
- **Bash** for Cluster Computing & parallel analyses (Slurm)
- **SQL** for database manipulation, **Git** for version control, **GitHub** for publishing reproducible code.